Compiled Solutions – Intermediate Division

1. **01-02 C1 Lisp Programming**

   (ADD (SUB 4 5) (ADD 6 3) (MULT 4 8)) simplifies to 40
   (ADD –1 9 32) = 40

2. **02-03 C1 Lisp Programming**

   (CDR (CAR (CDR '((b c) (a d f) (d))))) = (d f)
   (CDR (CAR '((a d f) (d)))) =
   (CDR '(a d f)) =
   (d f)

3. **03-04 C1 Lisp Programming**

   (MULT (ADD 2 3) (SUB 4 6)) = (MULT 5 -2) = -10 -10
   Note that (-10) is not a correct answer.

4. **04-05 C1 Lisp Programming**

   (EXP (DIV (MULT (ADD 2 (SUB 4 2)) 3) 2) 4) = 1296
   (EXP (DIV (MULT (ADD 2 2) 3) 2) 4) =
   (EXP (DIV (MULT 4 3) 2) 4) =
   (EXP (DIV 12 2) 4) =
   (EXP 6 4) = 1296

5. **05-06 C1 Lisp Programming**

   (DIV (MULT (ADD 2 3) (SQUARE 2)) (SUB (EXP 2 3) (ADD 2 1))) 4
   (ADD 2 3) = 5 (SQUARE 2) = 4 (EXP 2 3) = 8 (ADD 2 1) = 3
   (DIV (MULT 5 4) (SUB 8 3)) =(DIV 20 5) = 4

6. **06-07 C1 Lisp Programming**

   (DIV (MULT (ADD 1 4 5) (SUB 7 2)) (EXP 5 2)) = 2
   (EXP 5 2) = 25 (SUB 7 2) = 5 (ADD 1 4 5) = 10
   (DIV (MULT 10 5) 25) = (DIV 50 25) = 2

7. **07-08 C1 Lisp Programming**

   (ADD (ADD 3 4) (SUB 5 2) (MULT 3 2) (EXP 2 3)) 14
   = (ADD 3+4 5–2 3*2 2^3)
   = (ADD 7 3 6 8)
   = 24

8. **08-09 C1 Lisp Programming**

(DIV (ADD (ADD 3 4) (MULT 4 2)) (SUB 8 5)) 5
= (DIV (ADD 7 8) (SUB 8 5))
= (DIV 15 3)
= 5

9. **09-10 C1 Lisp Programming**

(SUB 9 5) = 9 -5 = 4 27
(EXP 2 3) = 2^3 = 8
(DIV 9 3) = 9/3 = 3
(MULT 3 5) = 3*5 = 15
So (ADD (SUB 9 5) (EXP 2 3) (MULT (DIV 9 3) 5))
= (ADD 4 8 15) = 27

10. **11-12 C2 Lisp Programming**

(ADD (SUB 5 2) (MULT 2 3) (DIV 9 3)) 12
(ADD 3 6 3) = 12

11. **11-12 C4 Lisp Programming**

(CAR (CAR (CDR (CDR '(a (b c) (a (b c))))))) a
= (CAR (CAR (CDR '((b c) (a (b c))))))
= (CAR (CAR '((a (b c)))))
= (CAR '(a (b c)))
= a

12. **12-13 C2 Lisp Programming**

(ADD (MULT 2 3) (EXP 3 2) (SUB 9 5) (DIV 8 4)) 21
= (ADD (MULT 2 3) (EXP 3 2) (SUB 9 5) 2)
= (ADD (MULT 2 3) (EXP 3 2) 4 2)
= (ADD (MULT 2 3) 9 4 2)
= (ADD 6 9 4 2)
= 21

13. **12-13 C4 Lisp Programming**

(CAR (CAR (CDR (CDR (CDR '(1 (2 3) 4 ((5 6) (7 8) 9))))))) (5 6)
= (CAR (CAR (CDR (CDR '((2 3) 4 ((5 6) (7 8) 9))))))
= (CAR (CAR (CDR '(4((5 6) (7 8) 9)))))
= (CAR (CAR '(((5 6) (7 8) 9))))
= (CAR '((5 6) (7 8)))
= (5 6)

14. **13-14 C2 Lisp Programming**

(ADD (MULT 2 5) (EXP 3 2) (SUB 4 (DIV 6 2))) 20
= (ADD (MULT 2 5) (EXP 3 2) (SUB 4 (DIV 6 2))))
= (ADD 2*5 3^2 (4 - 6/2)) = 10 + 9 + 1 = 20

15. **13-14 C4 Lisp Programming**

(CAR (CDR (CDR ‘(1 (2) (3 (4)))))) (3 (4))
= (CAR (CDR ‘((2) (3 (4)))))
= (CAR ‘((3 (4))))
= (3 (4))

16. **14-15 C2 Lisp Programming**

(ADD (ADD 2 3) (MULT 4 5) (SUB 6 2) (EXP 2 3)) 37
= (ADD 2+3 4*5 6-2 2^3)
= 5 + 20 + 4 + 8
= 37

17. **14-15 C4 Lisp Programming**

(CDR (CAR (CDR (CAR ‘((b (c d) (e f g)) (h (j k))))))) ‘(d) or (d)
= (CDR (CAR (CDR ‘(b (c d) (e f g)))))
= (CDR (CAR ‘((c d) (e f g))))
= (CDR ‘(c d)) = ‘(d)

18. **15-16 C2 Lisp Programming**

(MULT (ADD 2 3 1 (EXP 3 2)) (SUB (MULT 4 2) (DIV 10 2))) 45
= (MULT (ADD 2 3 1 9) (SUB 8 5))
= (MULT 15 3)
= 45

19. **15-16 C4 Lisp Programming**

(CDR (CAR (CDR (CDR ‘(a (b c) (d a (b c)) (a b)))))) (a (b c))
= (CDR (CAR (CDR ‘((b c) (d a (b c)) (a b)))))
= (CDR (CAR ‘((d a (b c)) (a b))))
= (CDR ‘(d a (b c)))
= (a (b c))